

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term (Regular) Examination, 2019
Department of Materials Science and Engineering

Transport Phenomena in Materials

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Deduce Bernoulli's equation. Through a refinery, fuel ethanol is flowing in a pipe at a velocity of 1m/s and a pressure of 101,300 Pa. The refinery needs the ethanol to be at a pressure of 202,600 Pa on a lower level. How far must the pipe drop in height in order to achieve this pressure? Assume that, the velocity does not change. (The density of ethanol is 789 kg/m³) (18)
- (b) Derive an expression for capillary rise of a liquid in a small tube dipped in it and prove that, the capillary rise is inversely proportional to the diameter of the tube. (10)
- (c) Determine the heat flux and heat transfer rate across an iron plate with area 0.5 m² and thickness 0.02m. [k= 70 W/ (m. °C)] when one of its surfaces is maintained at 60°C and the other at 20°C. (7)
- Q 2. (a) Consider a slab of thickness L as illustrated in fig: 2(a). A fluid at temperature $T_{\infty 1}$ with heat transfer coefficient h_1 flows over the surface at $x=0$, and another fluid at temperature $T_{\infty 2}$ with a heat transfer coefficient h_2 flows over the surface at $x=L$ of the plate. Develop an expression for the heat flow Q through an area A of the plate. Calculate that heat transfer rate through, $A= 1\text{m}^2$ of the slab for $T_{\infty 1}= 130\text{ }^\circ\text{C}$, $T_{\infty 2}= 30\text{ }^\circ\text{C}$, $h_1= 250\text{ W/(m}^2\cdot^\circ\text{C)}$ and $h_2= 500\text{ W/(m}^2\cdot^\circ\text{C)}$, $L= 4\text{ cm}$, $k= 20\text{ W/(m}\cdot^\circ\text{C)}$ (20)
- (b) A flat plate has one surface insulated and the other surface exposed to the sun. The exposed surface absorbs the solar radiation at a rate of 800 W/m² and dissipates it by both convection and radiation into ambient air at 300K. If the emissivity of the surface is $\epsilon= 0.9$ and the convection heat transfer coefficient between the plate and air is 12 W/ (m². °C), determine the temperature of the plate. (15)
- Q 3. (a) A hollow cylinder has inner radius r_1 and outer radius r_2 and length H, has a constant thermal conductivity k. If the inner surface is kept at temperature T_1 and outer surface at T_2 , show that the total heat transfer rate Q through the cylinder is given by- (18)

$$Q = \frac{T_1 - T_2}{R} W$$

Where, the thermal resistance R of the cylinder is- $R = \frac{\ln(\frac{r_2}{r_1})}{2\pi H k}$

- (b) An industrial furnace is made of fireclay brick of thickness 0.25m and thermal conductivity 1 W/ (m. °C). The outside surface is to be insulated with a material of thermal conductivity 0.05 W/ (m. °C). Determine the thickness of the insulation layer in order to limit the heat loss (17)

from the furnace wall to 1000 W/m^2 , when the inside surface of the wall is at 1030°C and the outside surface at 30°C .

- Q 4. (a) A 5cm thick iron plate [$k=60 \text{ W/(m}\cdot^\circ\text{C)}$, $C_p = 460 \text{ J/(kg}\cdot^\circ\text{C)}$, $\rho = 7850 \text{ kg/m}^3$ and $\alpha = 1.6 \times 10^{-5} \text{ m}^2/\text{s}$] is initially at 225°C . Suddenly, both surfaces are exposed to an ambient at 25°C with a heat transfer coefficient $500 \text{ W/(m}^2\cdot^\circ\text{C)}$, (18)
- Calculate the center temperature 2 minutes after the start of cooling.
 - Calculate the temperature at a depth 1 cm from surface 2 minutes after the start of cooling.
 - Calculate the energy removed from the plate per square meter during this time. [use the charts in fig: 4 if necessary]
- (b) Two 416 stainless steel plates each $L=1 \text{ cm}$ thick are bolted together. The thermal conductivity of the stainless steel can be taken as $k = 20 \text{ W/(m}\cdot^\circ\text{C)}$. The contact pressure is approximately 10 atm. Calculate the heat transfer rate across the plates for a temperature difference of $\Delta T = 100^\circ\text{C}$ per sq. m of surface, assuming that the contact conductance is $h_c = 4000 \text{ W/(m}^2\cdot^\circ\text{C)}$. (17)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) With a neat sketch, explain laminar and turbulent velocity boundary layers that may form on a flat plate. (17)
- (b) Water flows at a velocity 1 m/s over a flat plate of length 0.6 m . The water temperature is 350 K . In the laminar and turbulent regions, local convection coefficients are described by, $h_{\text{lam}}(x) = C_{\text{lam}}x^{-0.5}$; $h_{\text{turb}}(x) = C_{\text{turb}}x^{-0.2}$. Where x has units of meter. (18)
- At 350 K , $C_{\text{lam}} = 477 \text{ W/(m}^{1.5}\cdot\text{K)}$, $C_{\text{turb}} = 3600 \text{ W/(m}^{1.8}\cdot\text{K)}$. Determine the average convection coefficient over the entire plate.
- Q 6. (a) Derive a relationship for the logarithmic mean temperature difference for a single-pass parallel-flow heat exchanger. (17)
- (b) Engine oil at a mean temperature $T_1 = 80^\circ\text{C}$ and mean velocity $u = 0.1 \text{ m/s}$ flows inside a thin walled, horizontal copper tube with an ID of $D = 2.5 \text{ cm}$. The outer surface of the tube dissipates heat by free convection into atmospheric air at $T_\infty = 20^\circ\text{C}$. (18)
- Calculate the temperature of the tube wall, the overall heat transfer coefficient and the heat loss per unit length of the tube.
- Q 7. (a) Define solid angle. Deduce the relationship for blackbody emissive power with neat sketches. (12)
- (b) A surface with an area of 4 cm^2 emits radiation as a blackbody at 1000 K . (15)
- Calculate the radiation emitted into a solid angle subtended by $0 \leq \phi \leq 2\pi$ and $0 \leq \theta \leq \frac{\pi}{3}$
 - What fraction is the energy emitted into the solid angle at (i) of that emitted into the entire hemispherical space.
- (c) Derive Kirchhoff's law of radiation. (8)
- Q 8. (a) Consider an enclosure consisting of two parallel, infinite opaque plates. Surface 1 and 2 are gray, are kept at uniform temperatures T_1 and T_2 and have emissivities ϵ_1 and ϵ_2 and reflectivities ρ_1 and ρ_2 , respectively. (18)

- (i) Develop an expression for radiation exchange between the surfaces.
 - (ii) Calculate the net radiation flux q_{1-2} leaving plate 1 for $T_1=800\text{K}$, $T_2= 600\text{K}$, and $\epsilon_1 = \epsilon_2= 0.8$
- (b) Fused quartz transmits 85% of the thermal radiation at 1700K in the wavelength band (17) $\lambda_1 = 0.2$ to $\lambda_2 = 3 \mu\text{m}$ and is opaque to the radiation outside this range. If a blackbody radiation source at 1700K is placed in front of this fused quartz sheet, what is the rate of energy transmitted through a 0.1 sq. m area of the quartz sheet?

[see fig: 8(a) for radiation function chart]

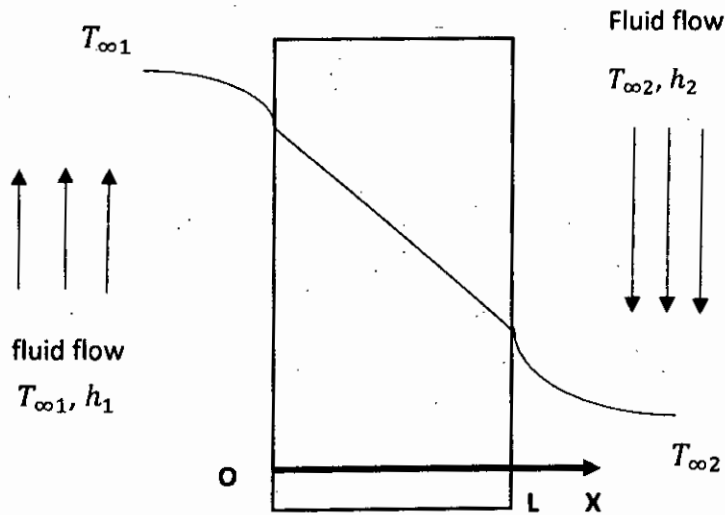
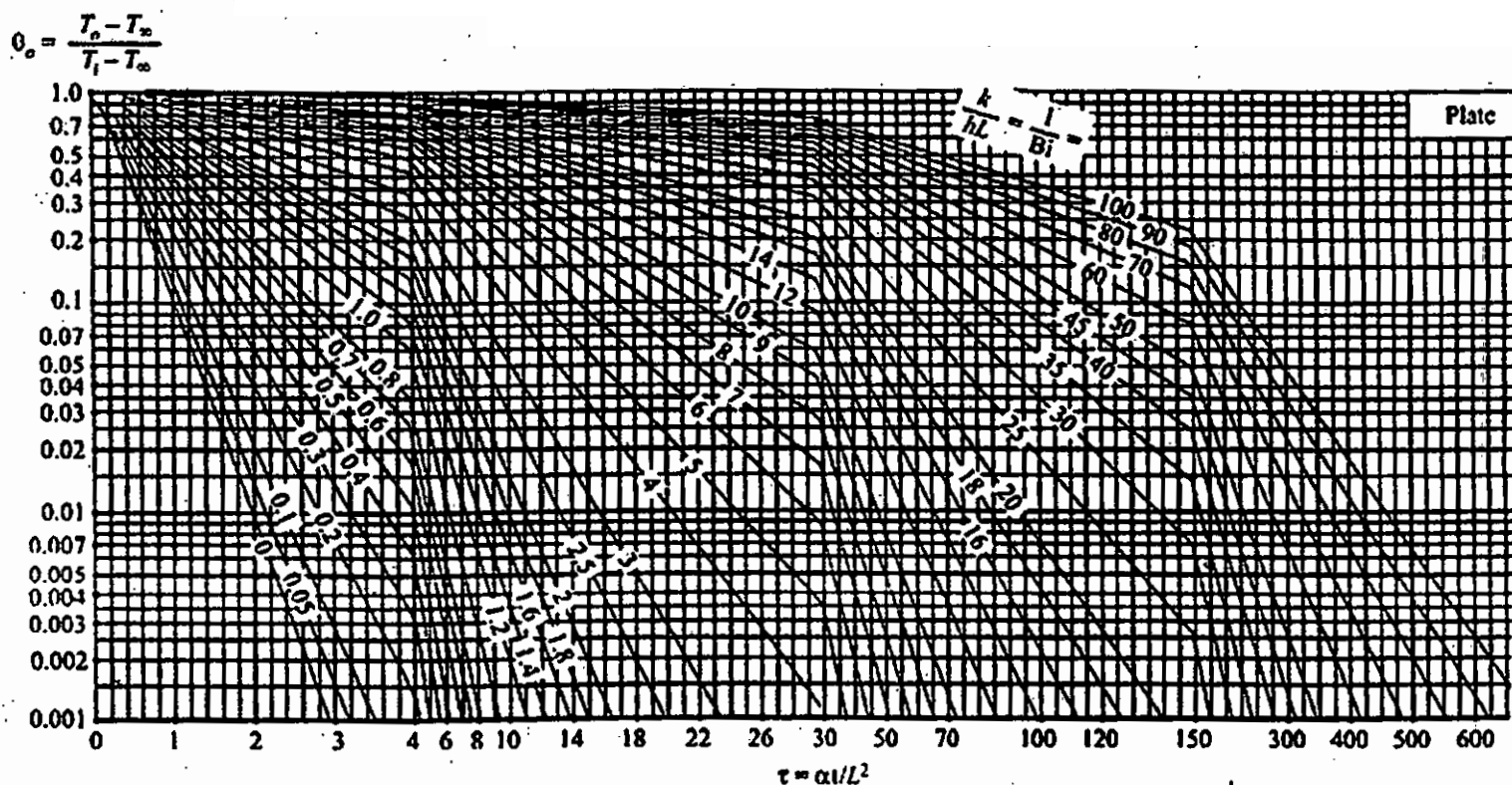


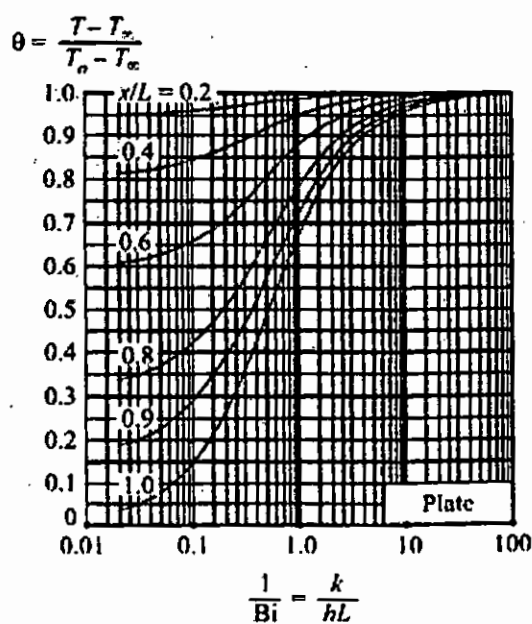
Fig: 2(a)

$\lambda T,$ $\mu\text{m} \cdot \text{K}$	f_λ	$\lambda T,$ $\mu\text{m} \cdot \text{K}$	f_λ
200	0.000000	6200	0.754140
400	0.000000	6400	0.769234
600	0.000000	6600	0.783199
800	0.000016	6800	0.796129
1000	0.000321	7000	0.808109
1200	0.002134	7200	0.819217
1400	0.007790	7400	0.829527
1600	0.019718	7600	0.839102
1800	0.039341	7800	0.848005
2000	0.066728	8000	0.856288
2200	0.100888	8500	0.874608
2400	0.140256	9000	0.890029
2600	0.183120	9500	0.903085
2800	0.227897	10,000	0.914199
3000	0.273232	10,500	0.923710
3200	0.318102	11,000	0.931890
3400	0.361735	11,500	0.939959
3600	0.403607	12,000	0.945098
3800	0.443382	13,000	0.955139
4000	0.480877	14,000	0.962898
4200	0.516014	15,000	0.969981
4400	0.548796	16,000	0.973814
4600	0.579280	18,000	0.980860
4800	0.607559	20,000	0.985602
5000	0.633747	25,000	0.992215
5200	0.658970	30,000	0.995340
5400	0.680360	40,000	0.997967
5600	0.701046	50,000	0.998953
5800	0.720158	75,000	0.999713
6000	0.737818	100,000	0.999905

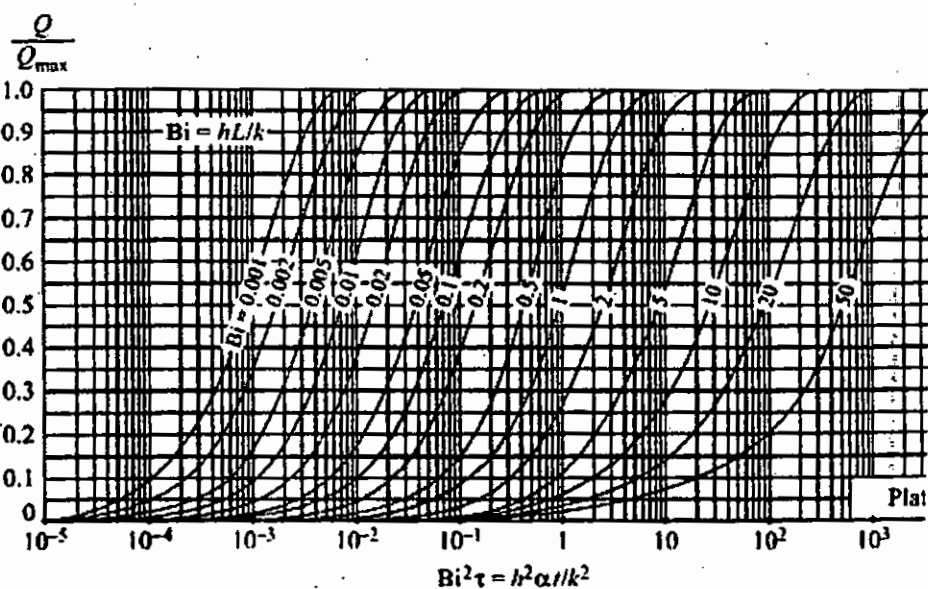
Fig: 8(a)- Blackbody radiation functions



(a) Midplane temperature



(b) Temperature distribution



(c) Heat transfer

Fig: 4 Transient temperature and heat transfer charts for a plane wall of thickness $2L$ initially at a uniform temperature T_i subjected to convection from both sides to an environment at temperature T_∞ with a convective coefficient of h .

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B.Sc. Engineering 3rd Year 1st Term (Regular) Examination, 2019
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MSE 3103
Materials Manufacturing Processes

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What do you understand by precipitation hardening? Explain the principles of precipitation hardening with necessary diagrams. 15
(b) What principles are involved in centrifugal casting? What types of objects are best made with this process. 10
(c) What sand molding process do you recommend for casting of frames, engine cylinders, rolls, large gears and housings? Justify your answer. 10
- Q 2. (a) Two gating designs for a mould 50cm x 30cm x 20cm are shown in figure 2(a) and (b). The cross sectional area of the gate is 6 sq. cm. Determine the filling time for both the designs. 15
(b) In case of rolling with smaller diameter rolls, larger diameter backup rolls are essential. Why? 10
(c) Explain the grain deformation in various mechanical processes. 10
- Q 3. (a) A strip with a cross-section of 150mm x 6mm is being rolled with 20% reduction of area, using 400mm diameter steel rolls. Before and after rolling, the shear yield stress of the material is 0.35 kN/mm² and 0.4 kN/mm² respectively. Calculate:
(i) The final strip thickness
(ii) The average shear yield stress during process
(iii) The angle subtended by the deformation zone at the roll center
(iv) The location of the neutral point θ_n . 20
Assume the coefficient of friction to be 0.1
- (b) With necessary diagrams, prove that, the tensile yield stress for plastic deformation of a metallic sample is $\sqrt{3}$ times the shear yield stress. 15
- Q 4. (a) A strip of lead with initial dimensions 24mm x 24mm x 150mm is forged between two flat dies to a final size of 6mm x 96mm x 150mm. If the coefficient of friction between the job and the dies is 0.25, determine the maximum forging force. The average yield stress of lead in tension is 7 kN/mm². 17
(b) Explain major casting defects and their reasons. 18

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) What do you understand by orthogonal cutting? For orthogonal cutting model, prove that,

$$\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}$$

18

where, all the notations indicate their usual meanings in machining.

- (b) In a machining operation that approximates orthogonal cutting, the cutting tool has a rake angle of 10° . The chip thickness before cut, $t_o = 0.5\text{mm}$ and the chip thickness after cut, $t_c = 1.125\text{mm}$. Calculate the shear plane angle and the shear strain in the operation. 10
- (c) Explain the formation of continuous chip with build up edge. 7

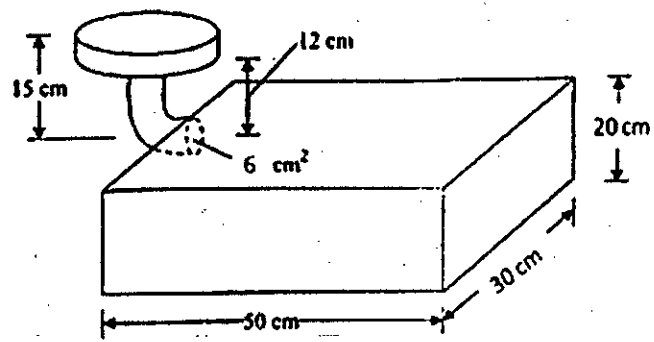
- Q 6. (a) The molecular weight data are exhibited in table 1. Compute the number average molecular weight and the weight average molecular weight. 10

Molecular Weight Range (g/mol)	x_i	w_i
8000-20000	0.05	0.02
20000-32000	0.15	0.08
32000-44000	0.21	0.17
44000-56000	0.28	0.29
56000-68000	0.18	0.23
68000-80000	0.10	0.16
80000-92000	0.03	0.05

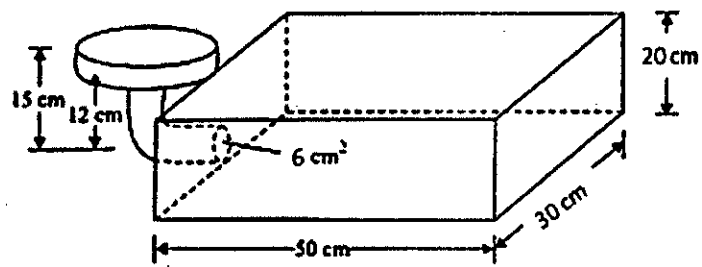
- (b) What do you understand by bambooing? Explain the reason of the formation of this defect with necessary diagrams. 12
- (c) Is it possible to extrude thermosetting plastics? Give appropriate reasons supporting your answer. 13
- Q 7. (a) Tempered glass has unique strength compared to traditional glass. How is the strength achieved? Briefly discuss the reasons with necessary diagrams. 12
- (b) With necessary chemical reactions and diagrams, briefly discuss how Portland cement is produced. 13
- (c) How does drying occur after slip casting? What are the necessary measures to achieve a high quality ceramic product.

- Q 8. (a) Write down the importance of powder metallurgy in the production of metallic parts. 10
- (b) What is the abbreviation of CLIP? How is this additive manufacturing (AM) technique better than the rest of AM techniques? 13
- (c) Which additive manufacturing route is the best for metallic element production? Explain. 12

Figure 2



(a)



(b)

Khulna University of Engineering & Technology
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MSE 3105
Welding and Materials Joining Process

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Classify different types of mechanical and metallurgical joining processes with brief description. (15)
- (b) Explain the basic operation of soldering and purposes of soldering flux. Write down the properties of soldering flux. (15)
- (c) What properties does a brazing filler metal must possess? (05)
- Q 2. (a) With appropriate diagrams, explain the types of weld joints. (10)
- (b) Name the essential components of a weld symbol. Explain the meanings of its basic components with suitable examples. (15)
- (c) Write short notes on: flame soldering and wave soldering. (10)
- Q 3. (a) Discuss the modes of metal transfer in gas-metal arc welding using suitable sketches. (15)
- (b) Explain the process of submerged arc welding with neat sketches. (15)
- (c) Compare electroslog welding process with submerged-arc welding process. (05)
- Q 4. (a) Describe gas-tungsten arc welding process with neat sketches. (15)
- (b) Discuss the working principle of electron beam welding with neat sketches. (15)
- (c) Write short note on the shielding gas used in Laser beam welding. (05)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Name three types of solid state welding process. Explain any of them. (15)
(b) Discuss the effects of welding speed on grain structure. (15)
(c) Sketch the microstructure that may form around a weld pool boundary. (05)
- Q 6. (a) What is melting efficiency? How can heat source efficiency be measured? (15)
(b) How can grain structure be controlled during welding? (10)
(c) What is distortion in welding? What are the remedies to reduce distortion in welds? (10)
- Q 7. (a) What are the factors that affect the crack susceptibility? Explain. (08)
(b) Briefly explain the characteristics of Inertia friction welding parameters. (15)
(c) What is micro-segregation in welding? Briefly describe the effects of dendrite tip undercooling on micro-segregation. (12)
- Q 8. (a) What are porosities? How do they form in the weld metal? How can they be avoided? (08)
(b) Explain the effects of welding parameters in heterogeneous nucleation. (15)
(c) Weld pool shape can be tear-drop or elliptical depending on welding speed. Based on this, evaluate development of columnar grain structure in the fusion zone of a welded metal at different welding speeds. (12)

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MSE 3107
Physical Metallurgy of Materials

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
 (ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Explain how the change of properties occurs after cold working of a metallic sample. (10)
 (b) Briefly explain the stages of stress relief process of a cold worked sample. (15)
 (c) Explain the hot working mechanism and also discuss how hot working is different from a cold working operation. (10)
- Q 2. (a) Draw a neat sketch of iron-iron carbide diagram with proper labelling and illustrate the microstructural changes during cooling from liquid state of the following samples with: (15)
 (i) Peritectic (ii) Eutectic and (iii) Eutectoid composition.
 (b) Discuss the effects of sulphur and phosphorous content in steel. (10)
 (c) To alter the critical cooling rate, changing the chemical composition is preferred rather than coarsening the austenitic grain- explain why. (10)
- Q 3. (a) Explain the mechanism of heat removal during quenching with necessary sketches. (15)
 (b) Write the names of the factor that affect the actual cooling rate during quenching process. Explain any ONE of them. (10)
 (c) Explain, why annealing should never be a final heat treatment process for a hypereutectoid steel. (10)
- Q 4. (a) It would seem more economical to base the material specification on the response to heat treatment (i.e. hardenability) rather than chemical composition- justify the statement. (15)
 (b) During annealing, why do we not follow upper critical line for hypereutectoid steel? (10)
 (c) Assume that, two different samples, A and B of similar composition is quenched in water and oil, respectively. What differences would you expect in their properties and why? (10)

(P.T.O.)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) What is Austempering? What are the principal advantages of austempering compared with the conventional quench and temper method? (15)
- (b) How can you get high hardness and low toughness in a steel by tempering process? Explain your answer with necessary diagrams. (10)
- (c) What are the limitations of Austempering and Martempering process? (10)
- Q 6. (a) Explain different modules of hardening of carburized parts. (15)
- (b) Differentiate between Pack Carburization and Gas Carburization. (10)
- (c) Illustrate the Nitriding process. Explain the microstructural changes that you expect from a nitriding operation. (10)
- Q 7. (a) Explain the process of Pack carburizing with necessary diagram. (15)
- (b) Why do we want to avoid hypereutectoid layer in the surface of a carburized sample? Explain in favor of your answer. (10)
- (c) How does white nitride layer form in low carbon steels? Explain how it affects the mechanical properties of steels. (10)
- Q 8. (a) What are the recommended practices of annealing for nodular cast iron? Explain along with their purposes. (15)
- (b) What are the purposes of heat treatment of nodular cast iron? (10)
- (c) Explain the effect of carbon and silicon in determining different types of cast iron. (10)

Khulna University of Engineering & Technology
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HUM 3127
Government and Sociology

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What do you mean by politics? Is there any benefit in politics? Give your reason. (10)
(b) What is state? Distinguish between the compulsory and the optional function of a modern welfare state. (15)
(c) What is government? What are the differences between state and government? (10)
- Q 2. (a) Describe the various forms of government. Which of them do you like the most? Why? (20)
(b) What is citizenship? What are the differences between a citizen and an alien? What are the qualities of a good citizen? (15)
- Q 3. (a) Explain the origin and evolution of state according to the social contract theory. (10)
(b) What is Marxism? State briefly the basic principles of Marxism. (15)
(c) Explain that, "Fascism is not based on doctrine but reality". (10)
- Q 4. (a) What is parliamentary and presidential form of government? Discuss the differences between parliamentary and presidential form of government. (15)
(b) What is socialism? What is the future of socialism? (10)
(c) What is constitution? What are the main characteristics of Bangladesh Constitution? (10)

(P.T.O.)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Define sociology. (05)
(b) Discuss the nature and scope of sociology. (15)
(c) Explain the emergence of sociology in the socio-cultural context. (15)
- Q 6. (a) What are the differences between society and community? (10)
(b) Explain the major role of institution and association. (10)
(c) Define social structure. Discuss the different elements of social structure. (15)
- Q 7. (a) What do you mean by scientific method? (05)
(b) Narrate different steps of scientific research. (15)
(c) What is socialization? How does culture influence on socialization and personality development of a human being? (15)
- Q 8. (a) What is social change? ⁰⁵~~(10)~~
(b) "Change is an inevitable part of society"- what do you think? Give example from your own society. (15)
(c) Find out the pattern of urbanization in third world countries. (15)